

Classical Mechanics

Lecture 5: Energy and the Hamiltonian

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Chapter 1

Energy and the Hamiltonian

Overview. We first sharpen the meaning of an active symmetry and review how translations and rotations generate momentum and angular momentum. The missing conservation law is energy. Its symmetry is translation in time, and its conserved quantity is the Hamiltonian $H = \sum_i p_i \dot{q}_i - L$. The derivation also shows exactly how explicit time dependence transfers energy between a chosen subsystem and its environment.

1.1 Questions Before the Formal Argument

We are approaching the absolute minimum of classical mechanics needed for the subjects that follow, though there is still much more mechanics worth learning. Two questions arose before the main argument began.

Question from the audience. Which books are useful after this introductory course? ^a

Response. The established advanced reference is Herbert Goldstein's *Classical Mechanics*. Cornelius Lanczos's book on the variational principles of mechanics is another relevant source. A class reading list is more useful than pretending that one lecturer can survey every available text.

^aLecture timestamp: 00:00:36–00:02:13.

The second question records a moment in the history of an unresolved measurement. In 2011 the OPERA experiment had reported an apparent early arrival of neutrinos sent from CERN to Gran Sasso. The reported discrepancy was shorter than the proton pulse used to create the neutrinos, so the timing analysis depended on comparing pulse profiles rather than tagging one neutrino at emission and recognizing the same particle at detection.

Question from the audience. Had the apparent faster-than-light neutrino result been resolved? ^a

Response. No conclusion was then justified. Beam-pulse shape, target heating, and synchronization all demanded technical scrutiny. One speculative proposal considered the motion of a GPS satellite: in the satellite frame the source and detector move while timing signals travel, producing corrections of order tens of nanoseconds. The estimate was interesting as an order-of-magnitude exercise, but it was not presented as a believable diagnosis of the experiment. Ordinary Lorentz contraction at the relevant speed was too small to supply the reported effect. The proper response was to wait for the experimental analysis.

^aLecture timestamp: 00:02:30–00:09:03.

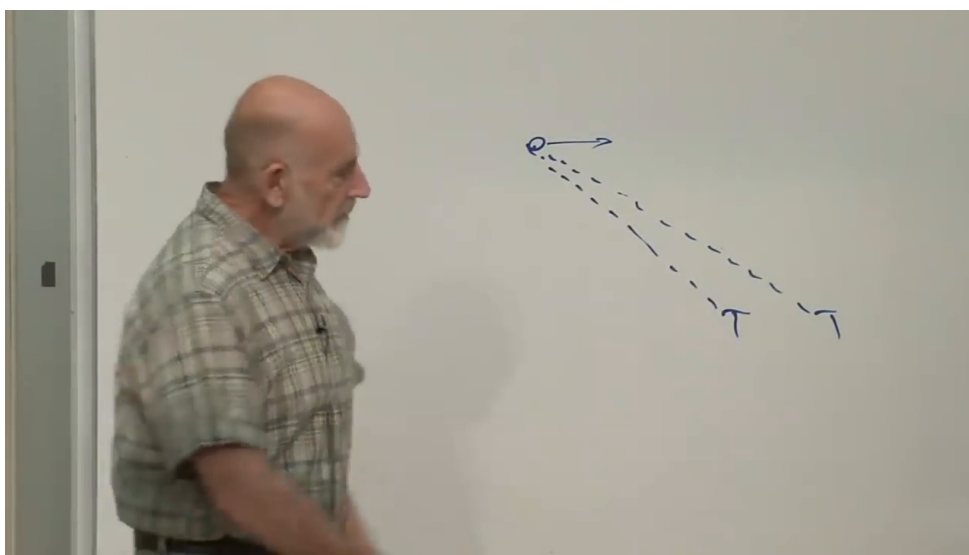


Figure 1.1: The satellite-frame timing sketch discussed during the historical neutrino-measurement question.

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The timing discussion is not needed for the mechanics below. It is useful nevertheless as a reminder that a plausible scale estimate is not the same as an established experimental explanation.

1.2 Active Symmetry and the Choice of System

Let us return to the point that was left too blurred in the preceding discussion: what precisely is a symmetry? Consider a particle on a line and the instruction

$$x \longrightarrow x + 1. \quad (1.1)$$

There are two ways to understand it. A *passive* transformation relabels the points of the line. A point once called $x = 0$ may now be called $x' = 1$, but the particle has not moved. An *active* transformation keeps the labels fixed and moves the particle itself from x to $x + 1$. Only the second operation changes its position relative to an external potential.

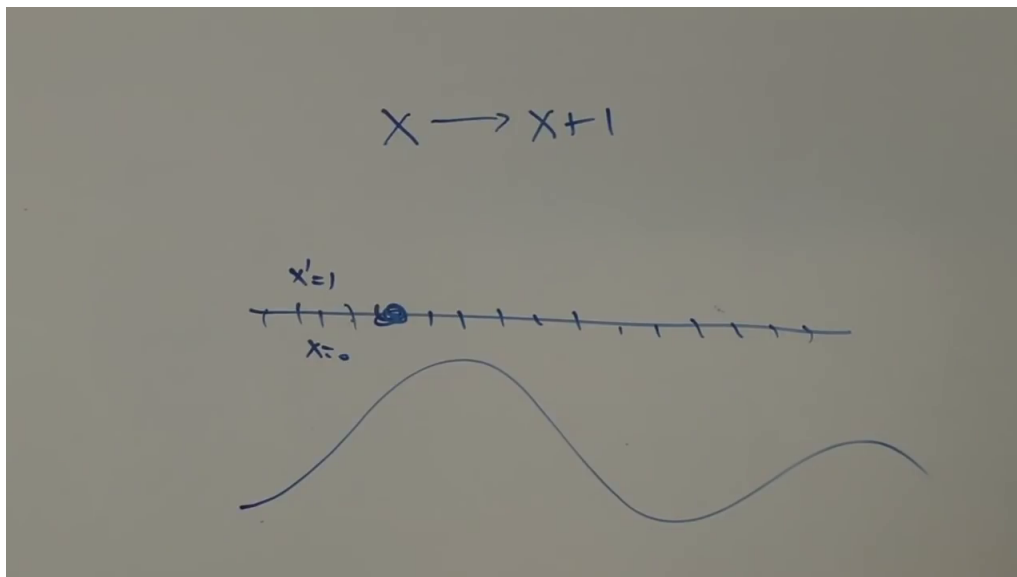


Figure 1.2: The rule $x \mapsto x + 1$, the fixed coordinate line, and an external potential distinguish active motion from passive relabeling.

Lecture frame: 00:11:20.

Question from the audience. If moving the particle changes its potential energy, is the source of that potential being excluded from the system? ^a

Response. Exactly. Calling the system “a particle in a potential” divides the world into the particle and a heavy environment, such as the Earth, that produces $V(x)$. The Earth does recoil, but its enormous mass makes that recoil negligible in this approximation. Moving the particle relative to the fixed Earth can change V ; translating the particle and Earth together would not.

^aLecture timestamp: 00:12:19–00:14:18.

Question from the audience. In the earlier rotation example, were the velocities changed as well as the positions? ^a

Response. Yes. A rigid active rotation acts on the entire history, not on one instantaneous position alone. Positions and velocity vectors are both rotated. By contrast, a constant translation applied at every time changes x but leaves $\dot{x}$ unchanged.

^aLecture timestamp: 00:14:26–00:15:29.

The distinction between system and environment will return twice: when a time-dependent apparatus exchanges energy with a subsystem, and when macroscopic friction hides energy in unobserved microscopic motion.

1.3 Infinitesimal Motions in Configuration Space

A configuration of a general mechanical system is a point in the space of generalized coordinates q_i . An active infinitesimal transformation moves every possible configuration to a nearby one. We write

$$\delta q_i = \epsilon f_i(q), \quad (1.2)$$

where the functions f_i specify one fixed displacement field on configuration space. They may depend on all the coordinates and some may vanish. The common factor ϵ is small; it separates the size of the step from the shape of the transformation.

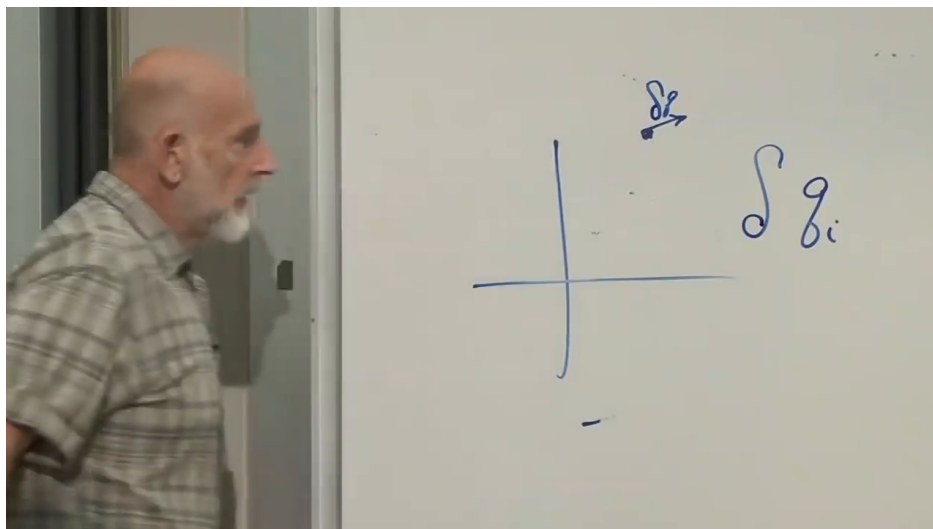


Figure 1.3: A displacement vector attached to each point of configuration space.

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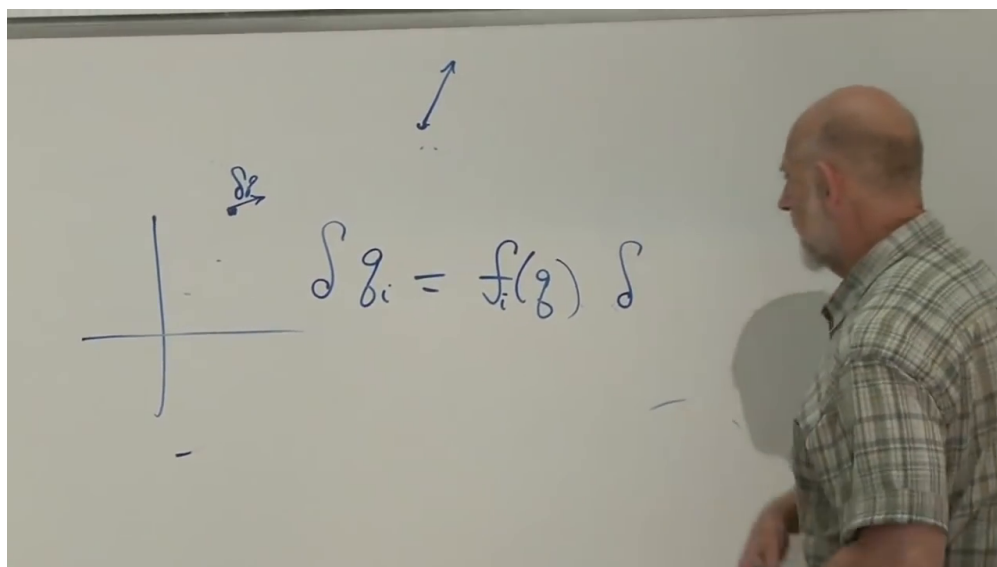


Figure 1.4: The coordinate-dependent infinitesimal transformation $\delta q_i = \epsilon f_i(q)$.

Lecture frame: 00:18:00.

The rule itself has no explicit time dependence. Its action on velocities is therefore

$$\delta \dot{q}_i = \frac{d}{dt}(\delta q_i) = \epsilon \sum_j \frac{\partial f_i}{\partial q_j} \dot{q}_j. \quad (1.3)$$

For a translation, $f_x = 1$, so $\delta \dot{x} = 0$. For a rotation, f_i depends on position, and the velocity changes.

Question from the audience. Is the final small factor in $\delta q_i = \epsilon f_i(q)$ the same constant at every point? ^a

Response. Yes. The single constant ϵ is a bookkeeping device that makes all coordinate changes first order in the same small quantity. The functions $f_i(q)$, not ϵ , contain the spatial dependence and need not themselves be small.

^aLecture timestamp: 00:19:58–00:20:51.

1.4 What Makes the Transformation a Symmetry?

An active transformation generally changes the potential energy. It can also change the kinetic energy. In curvilinear coordinates the kinetic term has the form

$$T = \frac{1}{2} \sum_{i,j} g_{ij}(q) \dot{q}_i \dot{q}_j, \quad (1.4)$$

so moving in configuration space can change its coefficients. The familiar $r^2 \dot{\theta}^2$ term in polar coordinates is the simplest example.

A transformation is a symmetry here when the *whole* Lagrangian has no first-order change,

$$\delta L = O(\epsilon^2), \quad (1.5)$$

and this must hold at every point of configuration space, not only at the configuration occupied by one particular solution.

For example, under $x \mapsto x + \epsilon$,

$$\delta V = \frac{\partial V}{\partial x} \epsilon. \quad (1.6)$$

If this vanishes for every possible x , then $\partial V / \partial x = 0$: the potential is independent of x . If it is unchanged under translations in every spatial direction, it is constant.

For a smooth one-parameter family, the vanishing first-order change at every point lets us compose infinitesimal steps into a finite symmetry.

Question from the audience. Must kinetic and potential energy remain invariant separately? ^a

Response. No. Only the total Lagrangian must be invariant. Some Lagrangians contain mixed coordinate–velocity terms and do not admit a useful separation into T and V , so separate invariance may not even be a meaningful question.

^aLecture timestamp: 00:24:44–00:25:33.

Question from the audience. Does “no change” mean exactly no change, or merely a change small enough to neglect? ^a

Response. For an infinitesimal transformation it means no term proportional to ϵ . Terms of order ϵ^2 are discarded in the local calculation. Because the first-order condition holds everywhere along the one-parameter orbit, repeating the local step yields the finite invariance.

^aLecture timestamp: 00:25:34–00:28:59.

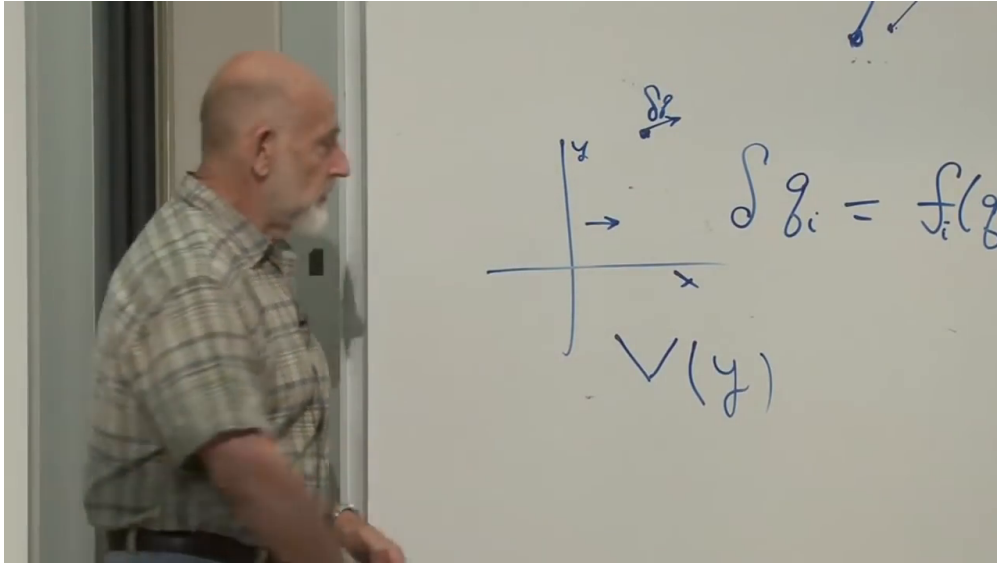


Figure 1.5: An x -translation tested against a potential $V(y)$ that is independent of x .

Lecture frame: 00:23:40.

1.5 Momentum and Angular Momentum Recovered

For the exact, time-independent point symmetries considered here, the result proved previously is

$$Q = \sum_i p_i f_i(q), \quad p_i = \frac{\partial L}{\partial \dot{q}_i}, \quad \dot{Q} = 0. \quad (1.7)$$

The factor ϵ does not belong in Q ; it measures only how far we move along the symmetry orbit.

For a translation in x , $f_x = 1$, and $Q = p_x$. Full translation invariance gives conservation of all three components of momentum. For many interacting particles, the symmetry translates every particle together, so

$$\mathbf{P} = \sum_a \mathbf{p}_a \quad (1.8)$$

is conserved when the interactions depend only on relative positions. Moving one particle relative to the rest is not the symmetry.

For an infinitesimal rotation in the x - y plane, choose the orientation

$$\delta x = -y\epsilon, \quad \delta y = x\epsilon. \quad (1.9)$$

It preserves the radius because

$$\delta(x^2 + y^2) = 2x\delta x + 2y\delta y = 0. \quad (1.10)$$

Differentiating Eq. (1.9) rotates the velocities as well. Thus $\dot{x}^2 + \dot{y}^2$ is invariant, and a central potential $V(\sqrt{x^2 + y^2})$ leaves the complete Lagrangian unchanged. The charge is

$$Q = -yp_x + xp_y = (\mathbf{r} \times \mathbf{p})_z. \quad (1.11)$$

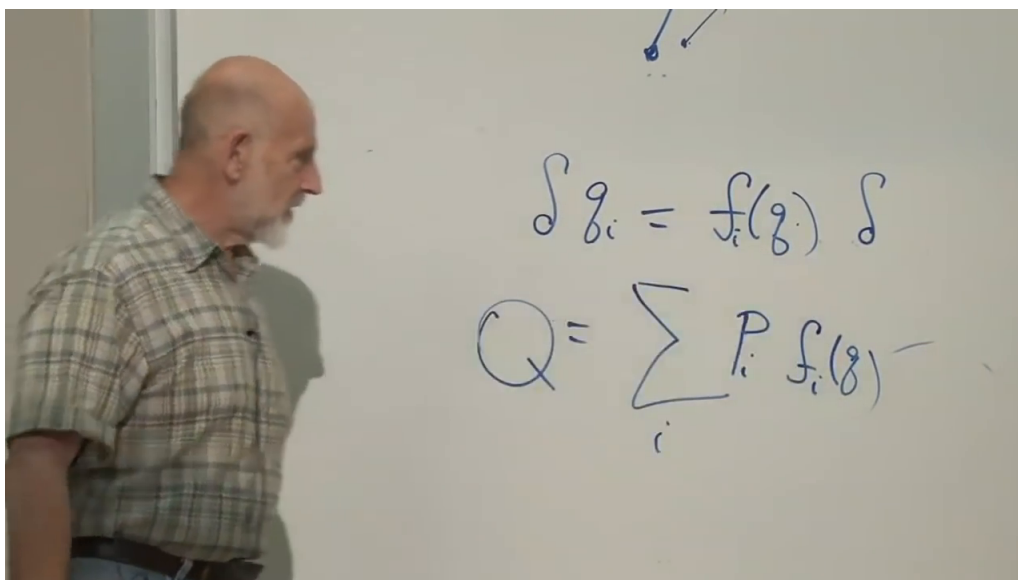


Figure 1.6: The general infinitesimal displacement and its conserved Noether charge.

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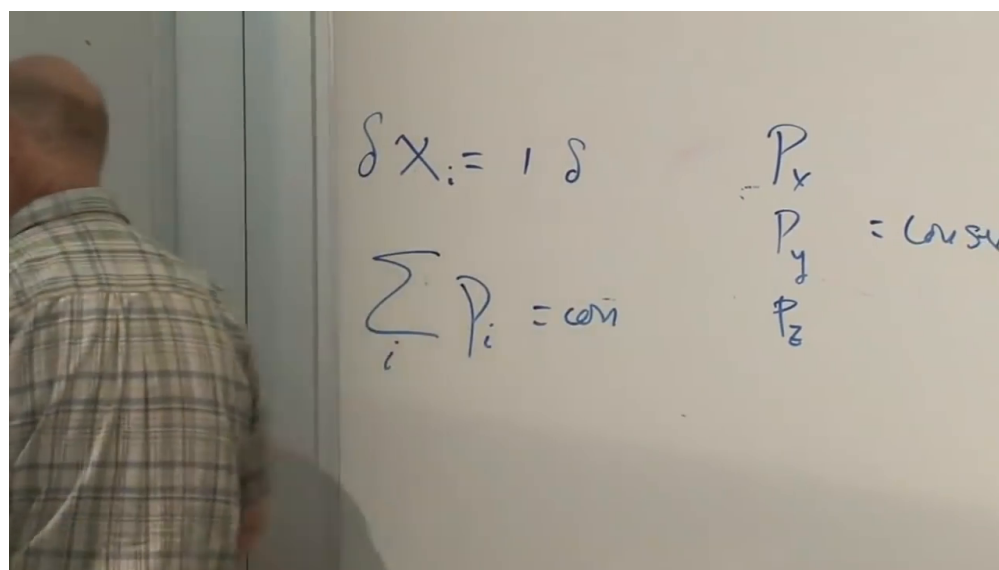


Figure 1.7: A common translation produces conservation of total momentum, component by component.

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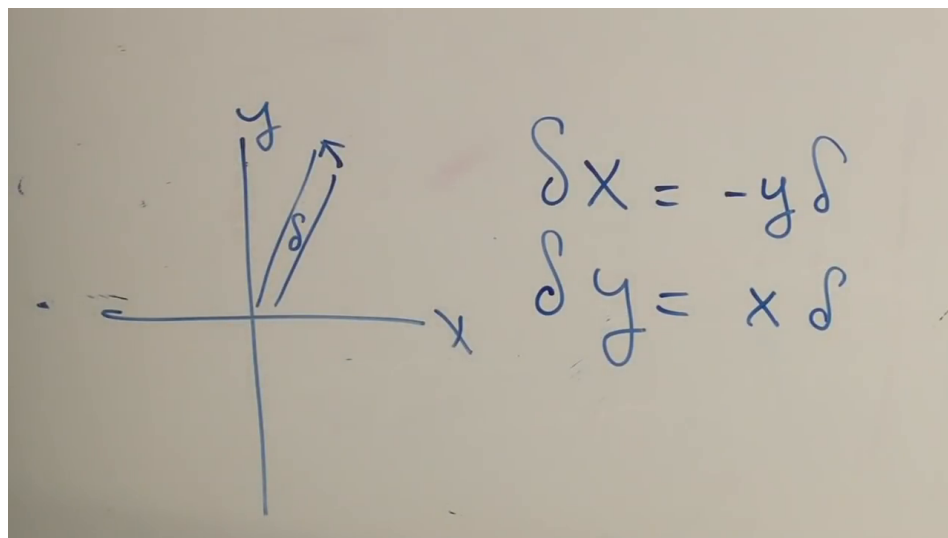


Figure 1.8: The geometry and coordinate formula for an infinitesimal planar rotation.

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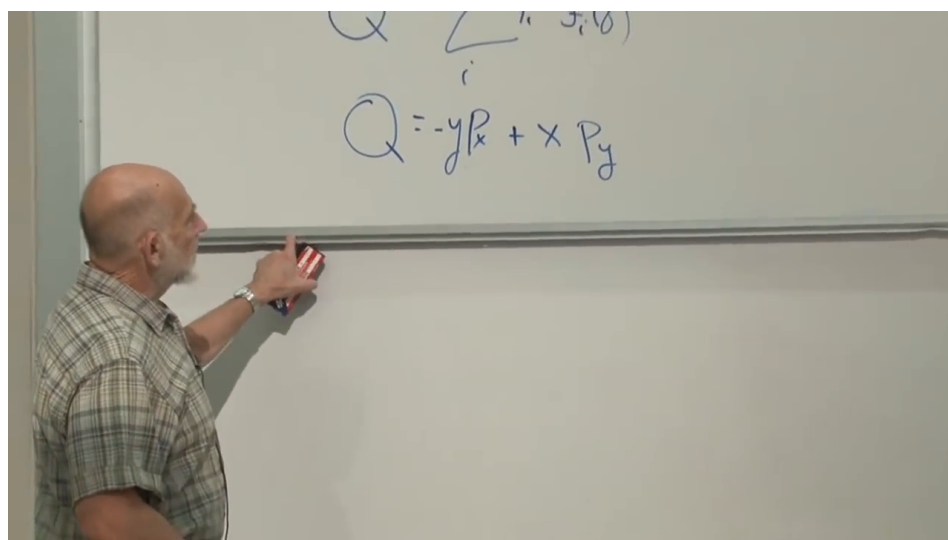


Figure 1.9: The rotational Noether charge, the z -component of angular momentum.

Lecture frame: 00:38:50.

Rotations in the other coordinate planes give the remaining components. For many particles the conserved vector is

$$\mathbf{J} = \sum_a \mathbf{r}_a \times \mathbf{p}_a. \quad (1.12)$$

A planet in a central stellar potential is the basic example: invariance under all rotations implies conservation of all components of angular momentum.

Question from the audience. What happens at a sharp cusp, where the derivative from one side differs from the derivative from the other? ^a

Response. That is a legitimate qualification. The present argument assumes differentiable functions with unique derivatives. An unstable equilibrium, such as an object balanced at a smooth maximum, is not itself a problem: the derivative there exists and vanishes. A literal cusp requires a modified treatment.

^aLecture timestamp: 00:42:25–00:44:21.

Question from the audience. If the transformation parameter is time independent, why does a rotation change the velocities? ^a

Response. The parameter is constant, but the displacement field is coordinate dependent. The universal rule is $\delta \dot{q}_i = d(\delta q_i)/dt$. A translation has constant δx , whereas a rotation has $\delta x = -y\epsilon$ and therefore $\delta \dot{x} = -\dot{y}\epsilon$.

^aLecture timestamp: 00:44:23–00:45:23.

1.6 Beyond the Elementary Kinetic-minus-Potential Form

In Cartesian coordinates for ordinary nonrelativistic particles without magnetic fields, $L = T - V$. More general coordinates produce the quadratic form of Eq. (1.4). A magnetic field introduces a term linear in velocity, schematically $\mathbf{A}(q) \cdot \dot{\mathbf{q}}$, which mixes coordinates and velocities. In such a case it is better to treat $L(q, \dot{q}, t)$ as one function than to force every term into conventional kinetic and potential boxes.

Question from the audience. Why might a Lagrangian fail to separate neatly into kinetic and potential energy, and is it still defined as $T - V$? ^a

Response. The obstruction may come from curvilinear coordinates or from genuine velocity-dependent interactions such as magnetism. Only when the Lagrangian consists of a quadratic velocity form minus a coordinate-only function do we name those pieces T and V . Otherwise the Lagrangian remains a perfectly good function of q and $\dot{q}$, but the split is not profitable. The correct general definition of energy will come from the Hamiltonian.

^aLecture timestamp: 00:45:26–00:48:13.

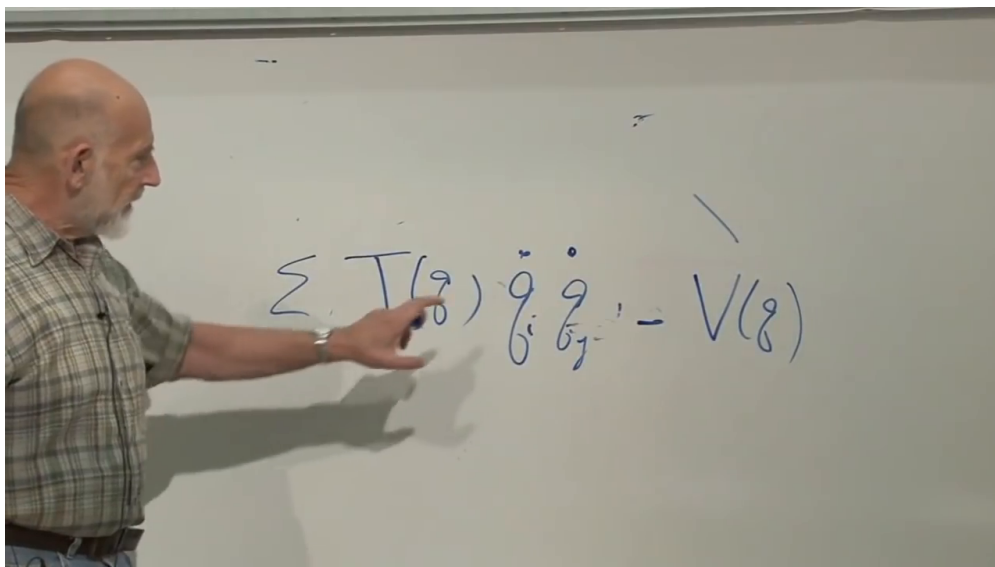


Figure 1.10: A general coordinate-dependent quadratic kinetic term beside a coordinate-only potential.

Lecture frame: 00:47:20.

Question from the audience. Was the hanging-chain example from the first meeting an application of least action? ^a

Response. It was an equilibrium problem governed by least energy, not a history selected by stationary action. For a static solution of an ordinary $T - V$ system, $\dot{p}_i = 0$ and the Euler–Lagrange equation reduces to $\partial V / \partial q_i = 0$, the condition for a stationary point of the potential.

^aLecture timestamp: 00:48:15–00:49:45.

Momentum and angular momentum have now been tied to spatial symmetries. The conspicuous missing quantity is energy.

1.7 Time Translation Invariance

Imagine an experiment that starts at t_0 , ends at t_1 , and produces an outcome. Repeat the same preparation later, starting at $t_0 + \Delta t$ and ending at $t_1 + \Delta t$. If the outcome is unchanged, the system is invariant under an active translation in time.

A charged particle between capacitor plates illustrates failure of this symmetry. With fixed plate charge, the same experiment can be performed at any time. If an external generator is steadily charging the plates, the particle sees a different electric field depending on when it is released. The particle is then an explicitly driven subsystem.

The same point appears in an oscillator,

$$L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}k(t)x^2. \quad (1.13)$$

An external electric or magnetic field might slowly alter the material and hence its effective spring constant. If k varies, shifting the oscillator experiment in time changes the result.

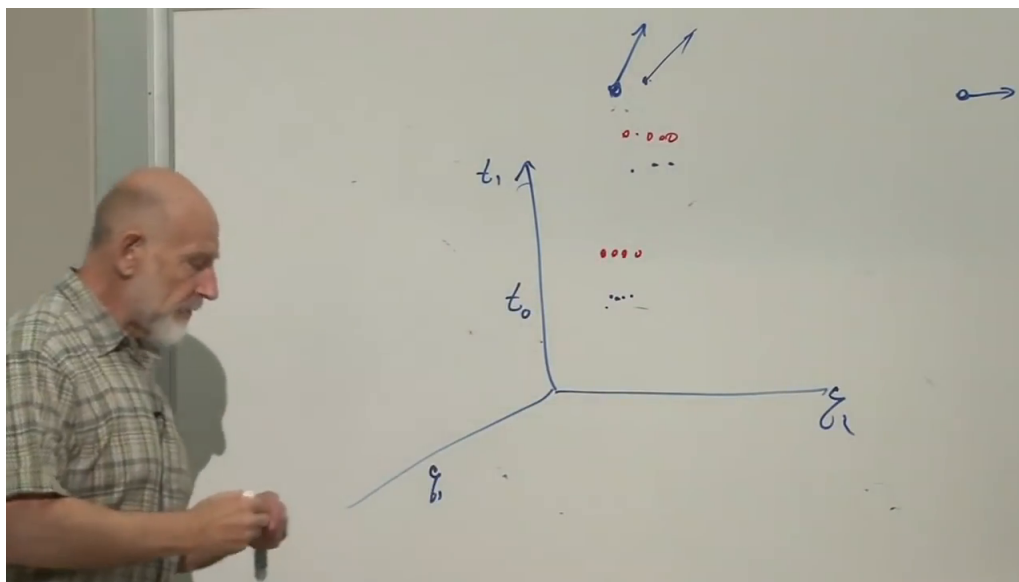


Figure 1.11: An experiment and its time-translated copy in configuration-space history.

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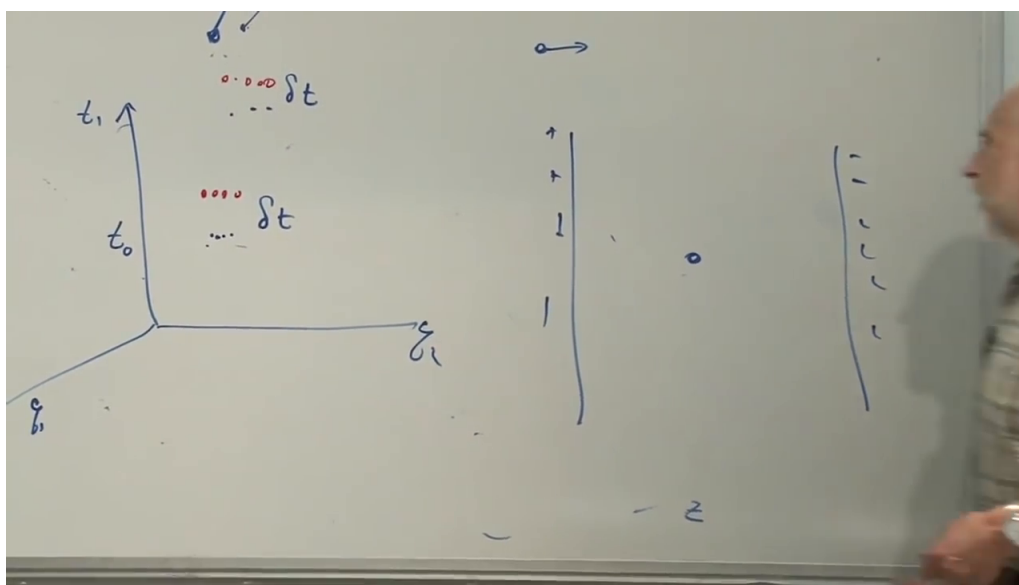


Figure 1.12: Fixed and externally changing capacitor plates contrasted beside the time-translation diagram.

Lecture frame: 00:54:45.

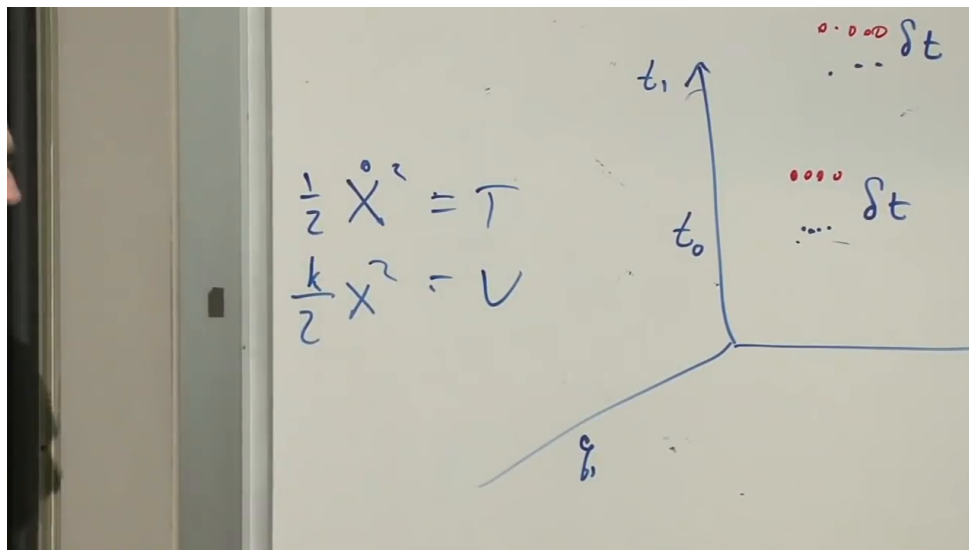


Figure 1.13: The oscillator's kinetic and potential terms during the discussion of a time-dependent spring constant.

Lecture frame: 00:56:45.

If the apparatus that changes k is included in the system, the full closed experiment can again be time translated and its full energy is conserved. The apparent nonconservation is energy exchange across a boundary we chose.

During the break the small transformation parameter was renamed ϵ , avoiding the visually confusing expression $\delta x = \delta$. A class recommendation for a book about Emmy Noether was also noted for the shared reading list.

Question from the audience. If the Lagrangian is not given as $T - V$, where does it come from? ^a

Response. It may be inferred from experiment, reconstructed from known equations of motion, guessed from symmetry and other theoretical principles, or found by trial and error. Once $L(q, \dot{q}, t)$ is given, the Euler–Lagrange calculus turns it into equations of motion. The formal argument does not require us to settle its origin first.

^aLecture timestamp: 01:00:17–01:02:24.

The operational test for time-translation invariance is now immediate. If knowledge of q and $\dot{q}$ suffices to determine L , without also knowing the clock time, then

$$\frac{\partial L}{\partial t} = 0. \quad (1.14)$$

This does not say that $L(q(t), \dot{q}(t))$ is constant along a motion. It says only that there is no time dependence beyond that carried by the evolving coordinates and velocities.

A rotating heavy dumbbell provides another driven example. If its prescribed motion creates a time-varying gravitational potential for a light test particle, the test-particle Lagrangian depends explicitly on time. Treating the dumbbell dynamically instead would enlarge the system and remove that external time dependence.

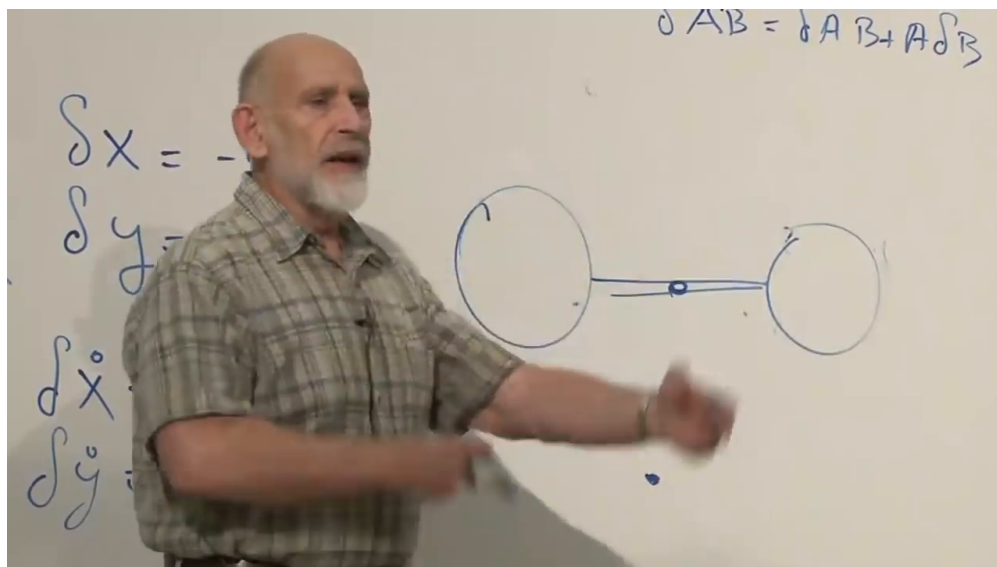


Figure 1.14: A prescribed rotating dumbbell acts as a time-dependent external source for a lighter particle.

Lecture frame: 01:06:25.

1.8 Deriving the Hamiltonian

Suppose we did not know the formula for energy and guessed that perhaps the Lagrangian itself were conserved. The direct test is to differentiate it along an actual trajectory. Begin, for simplicity, with no explicit time dependence:

$$\frac{dL}{dt} = \sum_i \left(\frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \right). \quad (1.15)$$

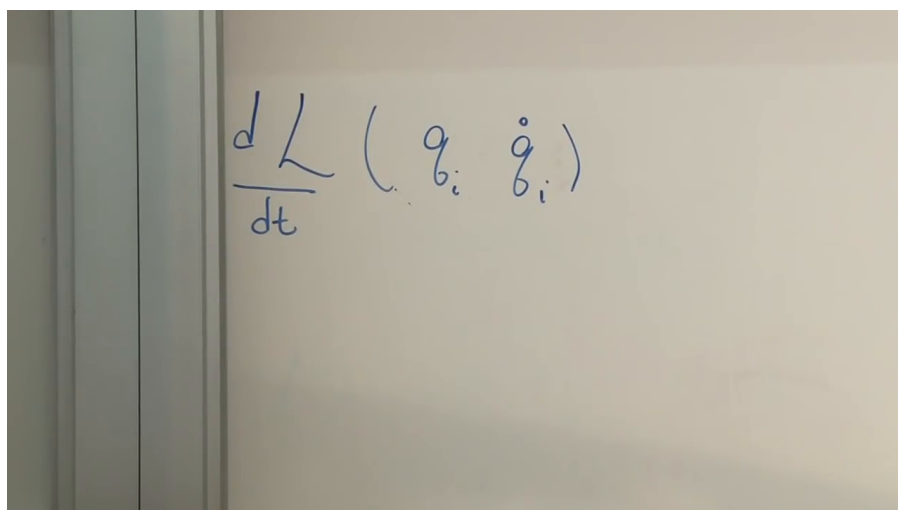


Figure 1.15: Beginning the derivative of $L(q_i, \dot{q}_i)$ along the trajectory.

Lecture frame: 01:08:51.

Question from the audience. Is a summation missing from the chain-rule expression? ^a

Response. Yes. Every coordinate and every velocity can change the Lagrangian, so both terms are summed over i .

^aLecture timestamp: 01:10:39–01:11:05.

Use the canonical momentum and the Euler–Lagrange equation,

$$p_i = \frac{\partial L}{\partial \dot{q}_i}, \quad \dot{p}_i = \frac{\partial L}{\partial q_i}. \quad (1.16)$$

Then Eq. (1.15) becomes

$$\begin{aligned} \frac{dL}{dt} &= \sum_i (\dot{p}_i \dot{q}_i + p_i \ddot{q}_i) \\ &= \frac{d}{dt} \sum_i p_i \dot{q}_i. \end{aligned} \quad (1.17)$$

The “magic” is only the product rule.

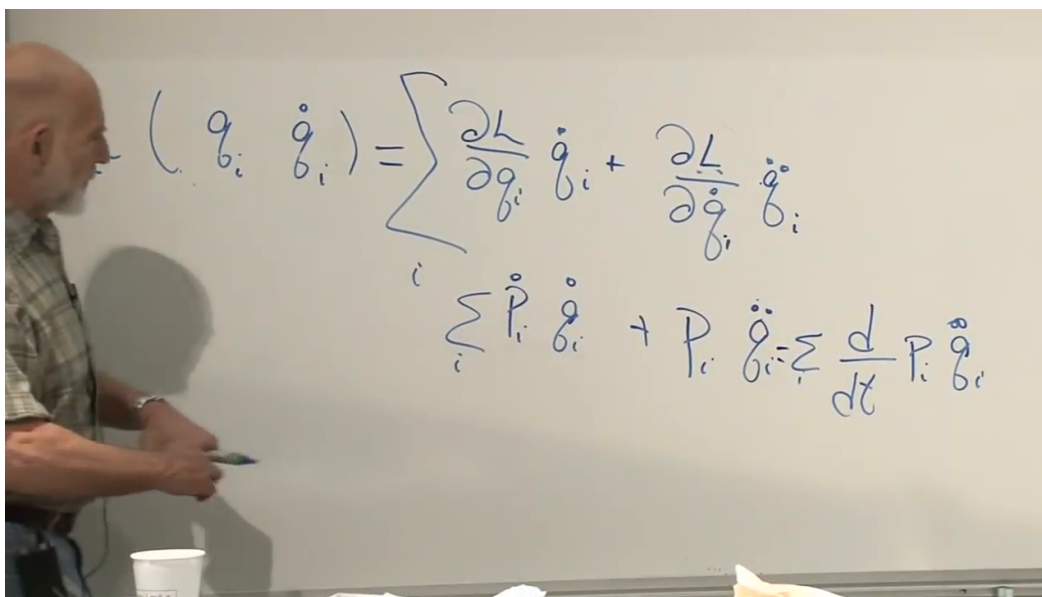


Figure 1.16: The chain rule converted, using the equations of motion, into the derivative of $\sum_i p_i \dot{q}_i$.

Lecture frame: 01:13:45.

Question from the audience. Should the final q_i in the product carry one dot or two? ^a

Response. Inside the product it is $p_i \dot{q}_i$, with one dot. Differentiating the product gives $\dot{p}_i \dot{q}_i + p_i \ddot{q}_i$, which explains the two dots in the expanded second term.

^aLecture timestamp: 01:14:31–01:15:32.

Subtracting the two derivatives gives a conserved combination. Historical convention chooses its negative and calls it the Hamiltonian:

$$H = \sum_i p_i \dot{q}_i - L. \quad (1.18)$$

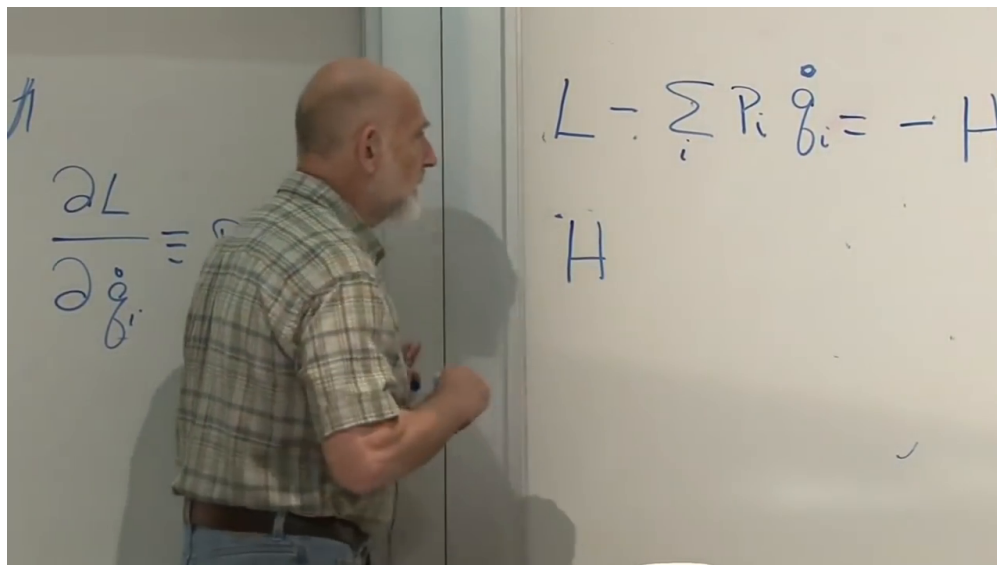


Figure 1.17: The conserved combination $L - \sum_i p_i \dot{q}_i = -H$ and the Hamiltonian definition.

Lecture frame: 01:18:05.

For one particle in a potential,

$$L = \frac{1}{2}m\dot{x}^2 - V(x), \quad p = m\dot{x}. \quad (1.19)$$

Therefore

$$\begin{aligned} H &= p\dot{x} - L \\ &= m\dot{x}^2 - \left(\frac{1}{2}m\dot{x}^2 - V(x) \right) \\ &= \frac{1}{2}m\dot{x}^2 + V(x) = T + V. \end{aligned} \quad (1.20)$$

The derivation did not assume a particular form of L . Thus Eq. (1.18), rather than $T + V$, is the general definition of energy in this formalism.

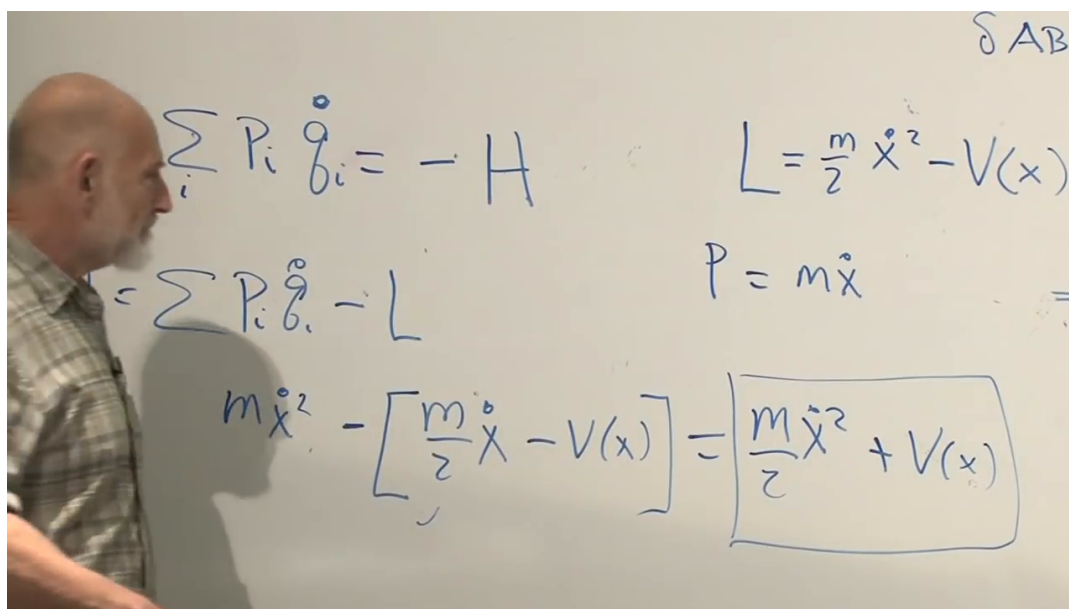
1.9 Explicit Time Dependence and Energy Exchange

Restore explicit time dependence in the chain rule:

$$\frac{dL}{dt} = \sum_i \left(\frac{\partial L}{\partial q_i} \dot{q}_i + \frac{\partial L}{\partial \dot{q}_i} \ddot{q}_i \right) + \frac{\partial L}{\partial t}. \quad (1.21)$$

The left side is the total derivative along the motion. The final partial derivative holds q and $\dot{q}$ fixed and differentiates only externally changing parameters. Repeating the same product-rule step yields

$$\boxed{\frac{dH}{dt} = -\frac{\partial L}{\partial t}}. \quad (1.22)$$



A man with a beard and a plaid shirt is pointing at a whiteboard. The whiteboard contains the following equations:

$$\sum_i p_i \dot{q}_i = -H$$

$$L = \frac{m}{2} \dot{x}^2 - V(x)$$

$$p = m \dot{x}$$

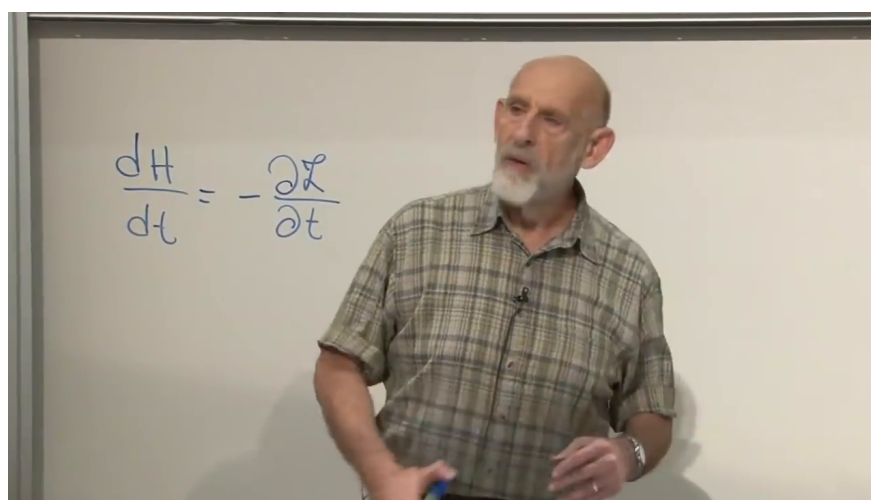
$$H = \sum_i p_i \dot{q}_i - L$$

$$m \dot{x}^2 - \left[\frac{m}{2} \dot{x}^2 - V(x) \right] = \boxed{\frac{m}{2} \dot{x}^2 + V(x)}$$

There is a small 'SAB' written in the top right corner of the whiteboard.

Figure 1.18: The Hamiltonian of a particle in a potential reduces to the familiar kinetic plus potential energy.

Lecture frame: 01:20:50.



A man with a beard and a plaid shirt is standing next to a whiteboard. The whiteboard contains the following equation:

$$\frac{dH}{dt} = -\frac{\partial \mathcal{L}}{\partial t}$$

Figure 1.19: The Hamiltonian changes only through explicit time dependence of the Lagrangian.

Lecture frame: 01:37:22.

For the driven oscillator of Eq. (1.13),

$$\frac{\partial L}{\partial t} = -\frac{1}{2}\dot{k}(t)x^2, \quad \dot{H} = \frac{1}{2}\dot{k}(t)x^2. \quad (1.23)$$

At $x = 0$ the instantaneous rate vanishes; away from the origin, changing the spring constant does work at a rate proportional to $x^2\dot{k}$.

Question from the audience. If energy is not conserved, is its definition the time derivative of the Hamiltonian? ^a

Response. No. The energy is H . Its derivative tells us how that energy changes. When the Lagrangian has the familiar form, $H = T + V$; when it does not, Eq. (1.18) supplies the definition. Enlarging the system to include the external driver restores conservation of the total Hamiltonian.

^aLecture timestamp: 01:27:11–01:28:04.

Question from the audience. How does this apply to the earlier observer on a rotating merry-go-round, where centrifugal and Coriolis terms appeared? ^a

Response. A steadily rotating coordinate system is a time-dependent change of variables, not a fixed spatial symmetry and not a simple translation in time. Its transformed Lagrangian can be used to construct a Hamiltonian, but that separate calculation is best postponed until magnetic forces are discussed, since the algebra has a closely related form.

^aLecture timestamp: 01:28:08–01:29:52.

1.10 The Lagrangian Toolkit in One View

The entire structure can now be written in a compact stack:

$$\begin{aligned} L &= L(q_i, \dot{q}_i, t), \\ p_i &= \frac{\partial L}{\partial \dot{q}_i}, \quad \dot{p}_i = \frac{\partial L}{\partial q_i}, \\ \delta q_i &= \epsilon f_i(q) \implies Q = \sum_i p_i f_i(q), \quad \dot{Q} = 0, \\ H &= \sum_i p_i \dot{q}_i - L, \quad \dot{H} = -\frac{\partial L}{\partial t}. \end{aligned} \quad (1.24)$$

These formulas contain a definition, equations of motion, the charge of an ordinary continuous symmetry, and the charge of time translation. There are few tricks; the important thing is to be able to rederive them rather than memorize disconnected expressions.

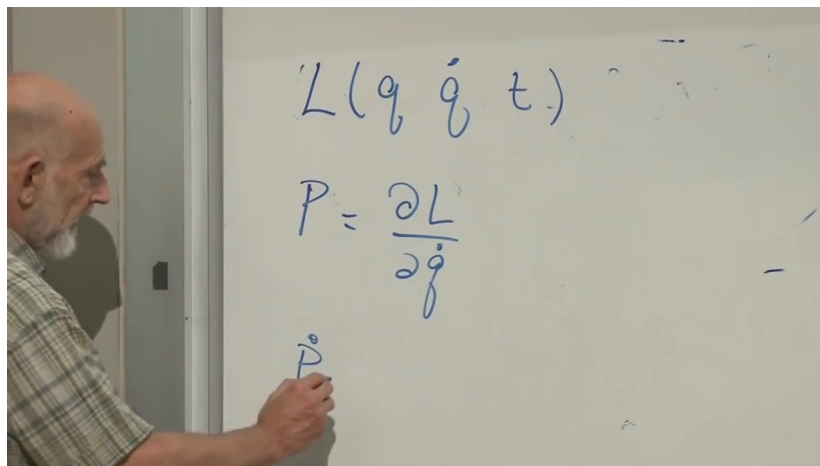


Figure 1.20: The board recap begins with $L(q, \dot{q}, t)$, canonical momentum, and the Euler–Lagrange equation.

Lecture frame: 01:31:37.

1.11 Why Use Lagrangian or Hamiltonian Mechanics?

Question from the audience. Can mechanics be described equally with L or H , and why choose one formulation? ^a

Response. They are equivalent when the Legendre transformation is regular, but they emphasize different structures. The Lagrangian begins naturally with stationary action. Hamiltonian mechanics is more abstract at first, but its equations are simple and it becomes central in quantum mechanics. The Hamiltonian is energy, but its role is much larger than the conservation law alone.

^aLecture timestamp: 01:37:16–01:39:18.

Question from the audience. Did Newton know the Hamiltonian form of energy conservation? ^a

Response. Not in this formulation, so far as the historical record indicates. Newton’s stated framework was the force law $F = ma$.

^aLecture timestamp: 01:39:29–01:39:56.

Question from the audience. Why should every physical system possess a Lagrangian? Is that merely experimental? ^a

Response. For the closed systems in view, it is an empirical fact and also follows from the deeper quantum description. Mathematically consistent differential equations can be invented that are not Euler–Lagrange equations. For example, $\dot{x} = -x$ is perfectly solvable, but as a fundamental closed one-coordinate law it has the information-losing character of dissipation. Classical mechanics does not logically precede quantum mechanics; its fundamental structure is inherited from quantum mechanics. The detailed reason must wait.

^aLecture timestamp: 01:39:57–01:42:37.

Question from the audience. If every real closed system has a Lagrangian, does every function we call a Lagrangian correspond to a system found in nature? ^a

Response. No. The formal calculus admits many functions that are not realized by any known physical system.

^aLecture timestamp: 01:42:39–01:42:54.

Question from the audience. Is the Lagrangian chiefly a tool for inventing potentials? ^a

Response. It is broader than that. It can organize candidate mechanical systems, and it is often the shortest practical route to the equations of motion.

^aLecture timestamp: 01:42:54–01:43:32.

The double pendulum displays that practical advantage. Choose two angles, compute the velocities of both masses in terms of those angles and their time derivatives, form $T - V$, and let the Euler–Lagrange equations handle the constraint forces implicitly. This is considerably cleaner than resolving all tensions and accelerations with $F = ma$.

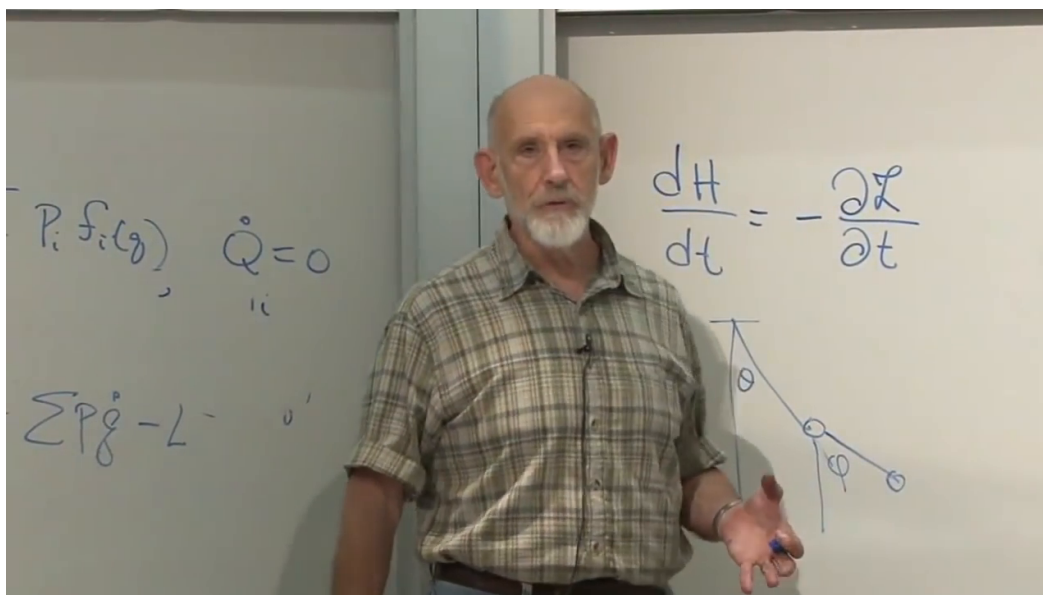


Figure 1.21: The energy-rate identity remains on the board beside the double-pendulum sketch.

Lecture frame: 01:45:39.

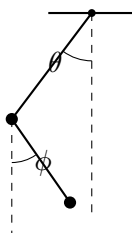


Figure 1.22: Generalized angular coordinates for a double pendulum.

Changing variables is similarly efficient: transform one Lagrangian and then derive its equations, rather than transforming every force equation separately.

Question from the audience. Is the double pendulum essentially another three-body system, such as Sun, Earth, and Moon, with one body treated as an anchor? ^a

Response. That analogy captures the hierarchy, although the rods impose different constraints. In either case a very massive or fixed component can be treated as the reference structure in an approximation.

^aLecture timestamp: 01:45:06–01:45:25.

Question from the audience. The double pendulum uses $T - V$, though more general Lagrangians were just emphasized. When is $T - V$ valid? ^a

Response. For ordinary nonrelativistic particles viewed in an inertial frame, with no magnetic velocity coupling, the familiar form applies. Rotating frames, relativistic systems, and magnetic fields require more general terms.

^aLecture timestamp: 01:46:08–01:46:58.

Question from the audience. Do collisions fall outside this description? ^a

Response. Elastic collisions arise from short-range forces and therefore from potentials. Inelastic collisions remain mechanical at the microscopic level, but energy enters internal vibrations, deformation, and heat. A center-of-mass description that omits those degrees of freedom is incomplete.

^aLecture timestamp: 01:47:00–01:48:05.

Question from the audience. Can a baseball struck by a bat be analyzed with $F = ma$? ^a

Response. Only at a coarse level. A complete account must follow deformation, heating, possible damage to the ball, and perhaps a breaking bat. Those internal degrees of freedom are why a realistic impact is much harder than a point-particle collision.

^aLecture timestamp: 01:48:05–01:48:33.

Question from the audience. What about an extended rigid body? ^a

Response. Rigid bodies can be studied either with Newtonian methods or with a $T - V$ Lagrangian written in translational and rotational coordinates. Constructing that Lagrangian is a useful exercise.

^aLecture timestamp: 01:48:34–01:49:00.

Question from the audience. Does the quantum energy–time uncertainty relation mean that energy is not strictly conserved? ^a

Response. No. Energy can be strictly conserved. The relation says that resolving energy with high accuracy requires observation over a long time; it does not replace the conservation law.

^aLecture timestamp: 01:49:01–01:49:29.

Question from the audience. Are Newtonian and Lagrangian mechanics logically equivalent?
^a

Response. They are equivalent within the ordinary domain where both formulations apply. One can nevertheless invent Lagrangians whose equations do not resemble the elementary form $m\ddot{x} = F(x)$.

^aLecture timestamp: 01:49:36–01:50:07.

1.12 A Coordinate-Dependent Kinetic Term

Consider the deliberately unfamiliar example

$$L = \frac{1}{2}q^2\dot{q}^2 - \frac{1}{2}q^2. \quad (1.25)$$

It is a valid Lagrangian with a coordinate-dependent kinetic coefficient. One could also consider velocity-linear terms such as $q\dot{q}$; in this one-dimensional example that particular term is a total derivative and would not alter the equations of motion.

The canonical momentum is

$$p = \frac{\partial L}{\partial \dot{q}} = q^2\dot{q}, \quad (1.26)$$

so

$$\dot{p} = q^2\ddot{q} + 2q\dot{q}^2, \quad \frac{\partial L}{\partial q} = q\dot{q}^2 - q. \quad (1.27)$$

The equation of motion is therefore

$$q^2\ddot{q} + q\dot{q}^2 = -q. \quad (1.28)$$

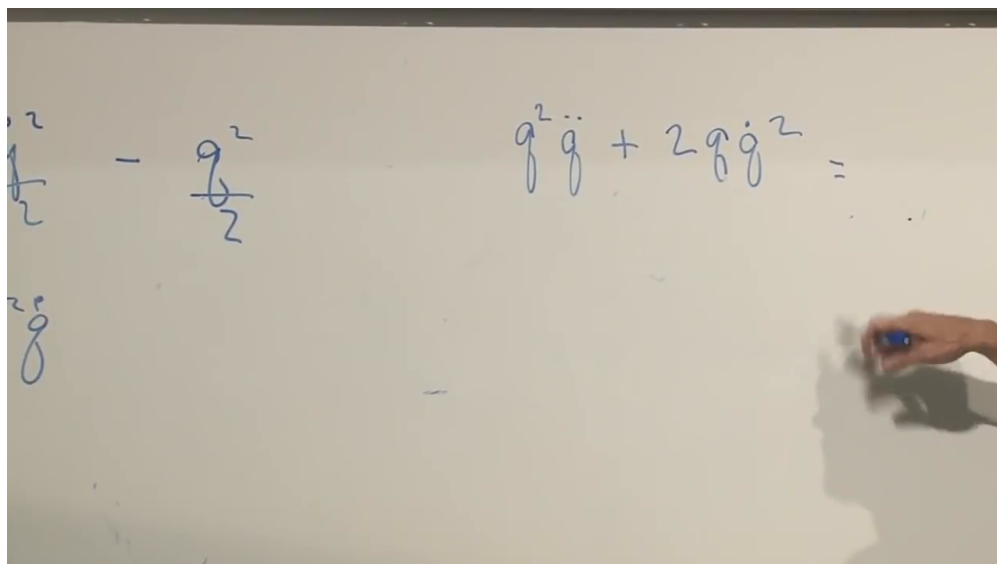


Figure 1.23: The differentiated momentum in the coordinate-dependent kinetic example.

Lecture frame: 01:52:32.

Question from the audience. Can the equation be divided by q ? ^a

Response. Only where $q \neq 0$. At $q = 0$ the kinetic coefficient and Legendre map are degenerate, so division would discard a singular point that needs separate treatment.

^aLecture timestamp: 01:53:22–01:53:39.

The Hamiltonian is

$$\begin{aligned} H &= p\dot{q} - L \\ &= q^2\dot{q}^2 - \left(\frac{1}{2}q^2\dot{q}^2 - \frac{1}{2}q^2 \right) \\ &= \frac{1}{2}q^2\dot{q}^2 + \frac{1}{2}q^2. \end{aligned} \tag{1.29}$$

Question from the audience. May $q^2\dot{q}^2/2$ still be called the kinetic energy? ^a

Response. Yes. In this worked example the Hamiltonian again looks like a positive kinetic term plus a potential term. The point is that the formal definitions worked without first assuming the elementary constant mass form.

^aLecture timestamp: 01:54:41–01:55:07.

1.13 Dissipation and the Deeper Origin of the Formalism

Question from the audience. How is friction included in a Lagrangian? ^a

Response. Not as a fundamental force in the simple closed finite-degree-of-freedom system considered here. Macroscopic friction is a statistical effect of enormous numbers of microscopic degrees of freedom. Those atoms still obey the underlying mechanics, but a reduced description that does not track them displays dissipation.

^aLecture timestamp: 01:55:09–01:56:17.

Question from the audience. How did Hamilton arrive at his formulation? ^a

Response. No historical derivation was offered here. The important task for the present course is to understand how the formulation works.

^aLecture timestamp: 01:56:22–01:56:37.

Question from the audience. Is there a more general recipe for obtaining a Lagrangian when it is not visibly $T - V$? ^a

Response. Experiment, an informed guess, a symmetry principle, or reconstruction from desired equations can supply the function. Asking the calculus itself where its input comes from is like asking the definition of a derivative where $f(x)$ came from. The calculus tells us what to do once the function is specified. Why nature's closed systems conform to this abstract structure is answered at a deeper level by quantum mechanics.

^aLecture timestamp: 01:56:43–01:58:54.

This point connects to the opening themes of the course. Fundamental Hamiltonian evolution is deterministic and reversible in the sense that trajectories do not merge and information is not simply

destroyed. The relation between that property and the phase-space formulation will be developed in stages.

Question from the audience. Did variational calculus merely work backward from the desired classical equations, and does quantum mechanics obtain its Lagrangian the same way? ^a

Response. In the elementary classical construction, one can indeed choose an action whose stationary paths reproduce the known equations. The logical order in quantum mechanics is different: it begins naturally with the Hamiltonian and can work toward a Lagrangian description. The Hamiltonian is more intuitive there, even though it first appears abstract in classical mechanics.

^aLecture timestamp: 01:59:37–02:01:07.

Question from the audience. Can quantum mechanics describe friction? ^a

Response. Yes, but a fundamental description must include the degrees of freedom of the environment. A particle moving through a viscous fluid cannot be described completely without the particles of the fluid. Friction is the complicated collective effect that emerges when those degrees of freedom are not followed individually.

^aLecture timestamp: 02:01:08–02:02:05.